Program : **Diploma in Chemical Engineering**	
Course Code : **6078**	Course Title : **Material Testing Lab (CH)**
Semester : **6**	Credits : **2.5**
Course Category : **Program Core**	
Periods per week : **3 (L: 0 T: 1 P: 3)**	Periods per semester : **60**

Course Objectives:

- To provide hands-on experience in the synthesis and testing of engineering materials.

Course Prerequisites:

Topic	Course code	Course name	Semester
Inorganic Chemicals		Inorganic Chemical Technology	3
Organic Chemicals		Organic Chemical Technology	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Prepare polymeric materials	22	Applying
CO2	Estimate the properties of polymers	23	Applying
CO3	Estimate the properties of ceramics	4	Applying
CO4	Demonstrate universal testing machine to estimate the strength of polymeric materials.	11	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2				3			
CO3				3			
CO4				2			

3-Strongly Mapped 2-Moderately Mapped 1- Weakly Mapped

Course Outline

Module Outcomes	Description	Duration (Hours)*	Cognitive Level
CO1	**Prepare polymeric materials**		
M1.01	Synthesize Phenol Formaldehyde Resin	5	Applying
M1.02	Synthesize Urea Formaldehyde Resin	5	Applying
M1.03	Synthesize Cuprammonium rayon	4	Applying
M1.04	Synthesize Poly Methyl Methacrylate	4	Applying
M1.05	Fabricate FRP using hand lay-up method.	4	Applying
CO2	**Estimate the properties of polymers**		
M2.01	Estimate the hardness of given polymer material using shore-D-hardness.	4	Applying
M2.02	Estimate softening point of given polymer material.	4	Applying
M2.03	Estimate the intrinsic viscosity of given polymer material using Ostwald viscometer.	4	Applying
M2.04	Estimate the specific gravity of given polymer material using hydrostatic bench method.	4	Applying
M2.05	Estimate the impact strength (Izod and Charpy) of plastic materials	4	Applying
	Lab Exam – I	3	

CO3	**Estimate the properties of ceramics**		
M3.01	Estimate the porosity of ceramic materials	4	Understandin g
CO4	**Demonstrate universal testing machine to estimate the strength of polymeric materials**		
M4.01	Estimate the tensile strength of given polymeric material using universal testing machine.	4	Understandin g
M4.02	Estimate the compressive strength of given polymeric material using universal testing machine.	4	Understandin g
	Lab Exam – II	3	

* Hours includes tutorial hour for each experiment + 3-hour laboratory experiment.

Text / Reference

T/R	Book Title/Author
T1	Outlines of Polymer Technology: Manufacture of Polymers; Sinha, R.
R1	Polymer Science and Technology; Fried, J.R
R2	An Introduction to Materials engineering and Science: for Chemical and Materials Engineers; B.S. Mittchell